

Úloha 6

Zadání

Vhodnou substitucí převed'te integrál na integrál z racionálních funkcí a ten se pokuste vyřešit:

$$\int \frac{x - \sqrt{x^2 + 3x + 2}}{x + \sqrt{x^2 + 3x + 2}} dx$$

Řešení

$$\int \frac{x - \sqrt{x^2 + 3x + 2}}{x + \sqrt{x^2 + 3x + 2}} dx$$

$$\sqrt{x^2 + 3x + 2} = -x + t$$

$$x = \frac{t^2 - 2}{3 + 2t}$$

$$dx = \frac{2t^2 + 6t + 4}{(3 + 2t)^2} dt$$

$$\int \frac{x - \sqrt{x^2 + 3x + 2}}{x + \sqrt{x^2 + 3x + 2}} dx = \int \frac{\frac{t^2 - 2}{3 + 2t} - t}{\frac{t^2 - 2}{3 + 2t} + t} \frac{2t^2 + 6t + 4}{(3 + 2t)^2} dt =$$

$$\int \frac{(-3t - 4)(2t^2 + 6t + 4)}{t(2t + 3)^3} dt = -2 \int \frac{3t^3 + 9t^2 + 6t + 4t^2 + 12t + 8}{t(2t + 3)^3} dt =$$

$$-2 \int \frac{3t^3 + 13t^2 + 18t + 8}{t(2t + 3)^3} dt = -2 \int \frac{A}{t} + \frac{B}{2t + 3} + \frac{C}{(2t + 3)^2} + \frac{D}{(2t + 3)^3} dt$$

$$A(2t+3)^3 + Bt(2t+3)^2 + Ct(2t+3) + Dt = 3t^3 + 13t^2 + 18t + 8$$

$$t = 0 \quad A = \frac{8}{27}$$

$$t = \frac{-3}{2} \quad D = \frac{-1}{12}$$

$$\frac{64}{27}t^3 + 4Bt^3 = 3t^3$$

$$B = \frac{17}{108}$$

$$\frac{32}{3}t^2 + \frac{17}{9}t^2 + 2Ct^2 = 13t^2$$

$$C = \frac{2}{9}$$

$$\begin{aligned} & -2 \int \frac{8}{27t} + \frac{17}{108(2t+3)} + \frac{2}{9(2t+3)^2} - \frac{1}{12(2t+3)^3} dt = \\ & -\frac{16}{27} \ln |t| - \frac{17}{108} \ln |2t+3| + \frac{2}{9(2t+3)} - \frac{1}{24(2t+3)^2} + C = \\ & -\frac{16}{27} \ln |t| - \frac{17}{108} \ln |2t+3| + \frac{32t+45}{72(2t+3)^2} + C \end{aligned}$$

Výsledek

$$\begin{aligned} & -\frac{16}{27} \ln |\sqrt{x^2+3x+2}+x| - \frac{17}{108} \ln |2\sqrt{x^2+3x+2}+2x+3| + \\ & + \frac{32\sqrt{x^2+3x+2}+32x+45}{72(2\sqrt{x^2+3x+2}+2x+3)^2} + C \end{aligned}$$